Investigation of the Influence of Sunspot Cycles on Temperature Variability in New York City (1948–Present): A Preliminary Wavelet Analysis Approach

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Abstract:

This study represents an initial exploration into the relationship between solar activity and temperature variability in New York City, using historical data spanning from 1948 to the present. In this early-stage investigation, sunspot and temperature records were downloaded, merged, and normalized to ensure consistency across the datasets. Wavelet analysis was then applied to extract variable frequency components from both time series.

Preliminary findings indicate that the sunspot data prominently exhibits an approximately 11-year cycle as the dominant signal, alongside several minor periodicities that are partially obscured by this primary cycle. In contrast, the temperature data reveal a strong annual cycle with an exact 365-day periodicity, reflecting the regularity of seasonal variations in the region.

In addition to identifying these cyclic behaviors, this introductory research integrates a geohazard perspective by investigating when and how peak intensities in these variables might signal heightened risk. By correlating periods of intensified solar activity and temperature anomalies with historical geohazard occurrences, the study lays the groundwork for a forecasting framework that could eventually predict the timing of extreme weather events, urban heat stresses, and other climate-related hazards.

Future work will focus on advanced noise reduction techniques to isolate additional peaks in both datasets, followed by a detailed coherence analysis to further refine predictive capabilities for geohazard risk. This preliminary analysis serves as a stepping stone toward more comprehensive investigations into solar-climate interactions and their implications for urban resilience.

Background of Study:

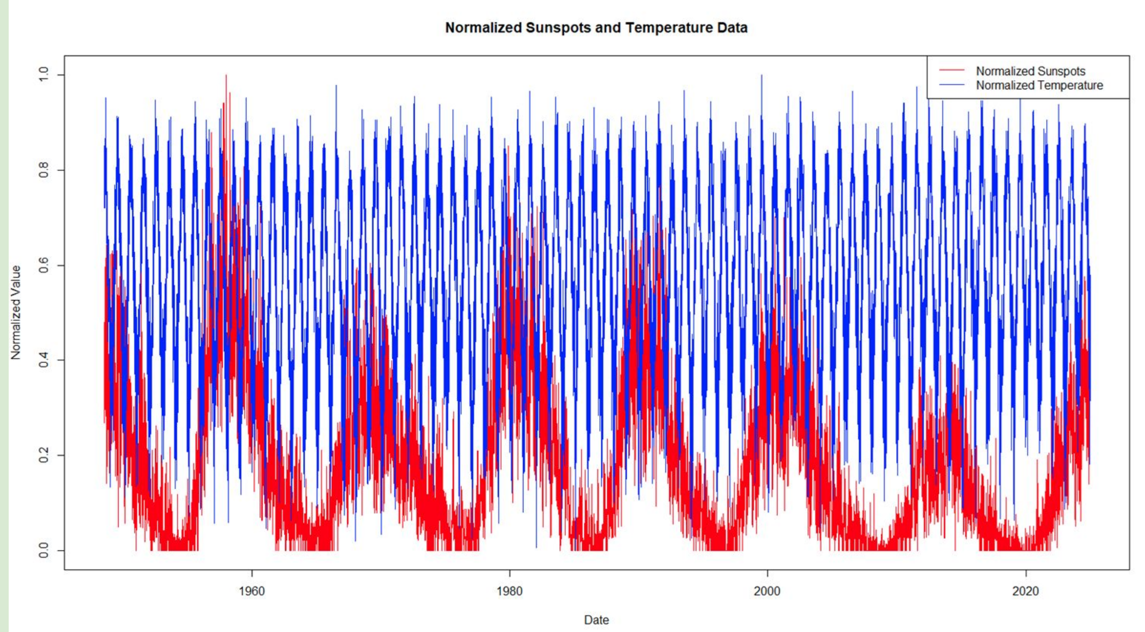
The Sun plays a central role in Earth's climate system, emitting energy that drives atmospheric circulation, weather, and temperature regulation. One key indicator of solar variability is the **sunspot cycle**, an approximately 11-year oscillation in solar magnetic activity. Sunspots are darker, cooler areas on the solar surface that fluctuate in number and intensity over time. Scientific interest has grown around whether these solar cycles influence climate variables, particularly temperature patterns, on regional and global scales.

While global temperature anomalies and long-term climate change are primarily attributed to greenhouse gas emissions, natural forcings like solar activity are still relevant for understanding short- and medium-term climatic variability. Previous research has suggested potential correlations between sunspot maxima and periods of slightly elevated global temperatures, but regional studies are less common, especially those focused on localized urban settings such as New York City and Long Island.

This study investigates whether fluctuations in sunspot activity correlate with changes in average temperature patterns in Islip, NY, over the period 2002–2024. By applying wavelet analysis, time-series smoothing (KZ filter), and normalization techniques in R, the project aims to determine if solar cycles leave detectable imprints on regional temperature variability. Through comparison of normalized sunspot and temperature trends and their frequency power spectra, this research contributes to the broader understanding of solar-climate interactions at the urban scale.

Method/Materials:

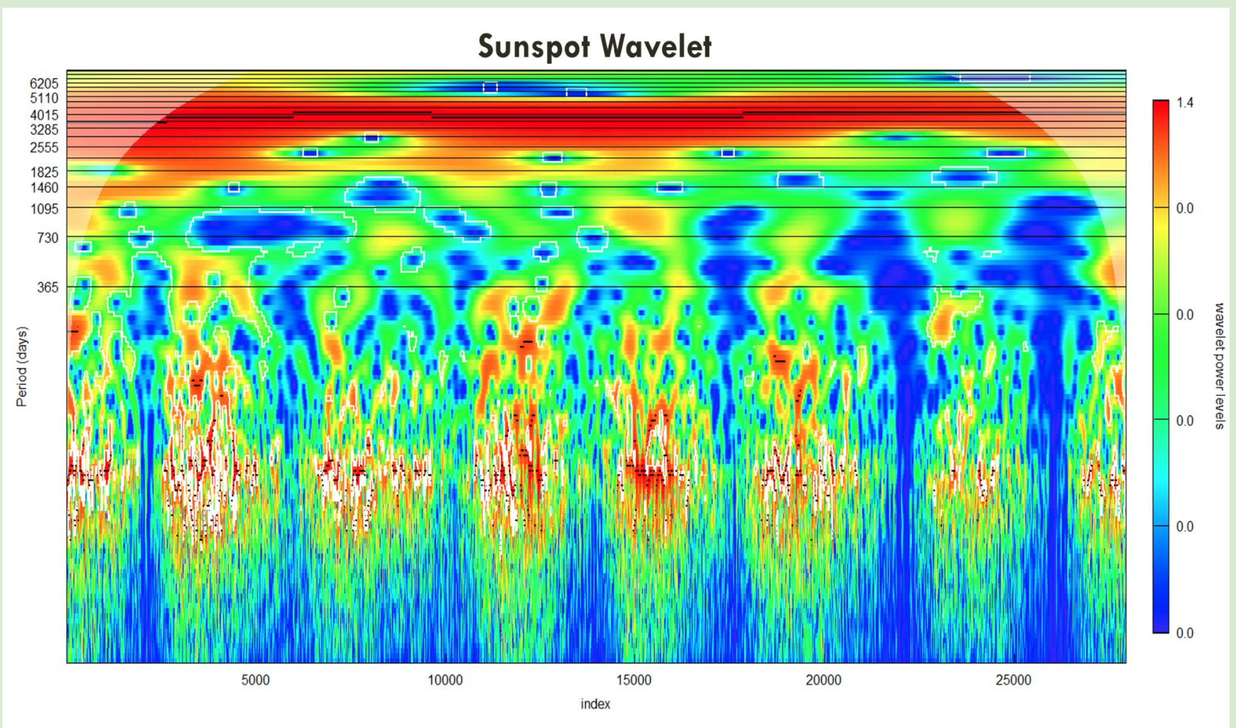
- **analyze both solar and climatological time-series data to investigate potential links between sunspot cycles and temperature patterns in the New York City and Long Island region.**
- **The Monthly sunspot Time-series were sourced from "`sunspot.month`" dataset adn was visually plotted**
- **The yearly average temperature time-series (from 1/1/02 to 12/31/24) was obtained from the islip macarthur station**
- **Both sunspot and temperature time-series were plotted and noramized**
- **Wavelet analysis were both performed for both time-series**



Results After Normalizing Both Data Sets

- Average temperatures tend to increase slightly following sunspot activity peaks, indicating a delayed thermal response.
- Temperature cycles remain seasonally consistent, but sunspot peaks correspond with sharper fluctuations, highlighted in red on the normalized graph.
- Periods of high sunspot activity are generally associated with above-average temperatures, suggesting a weak but observable solar influence.
- Years with minimal sunspot activity tend to have slightly lower average temperatures, though the effect is modest.
- The time lag between sunspot peaks and temperature changes suggests solar activity may modulate Earth's energy balance indirectly, warranting further lagged-correlation analysis.

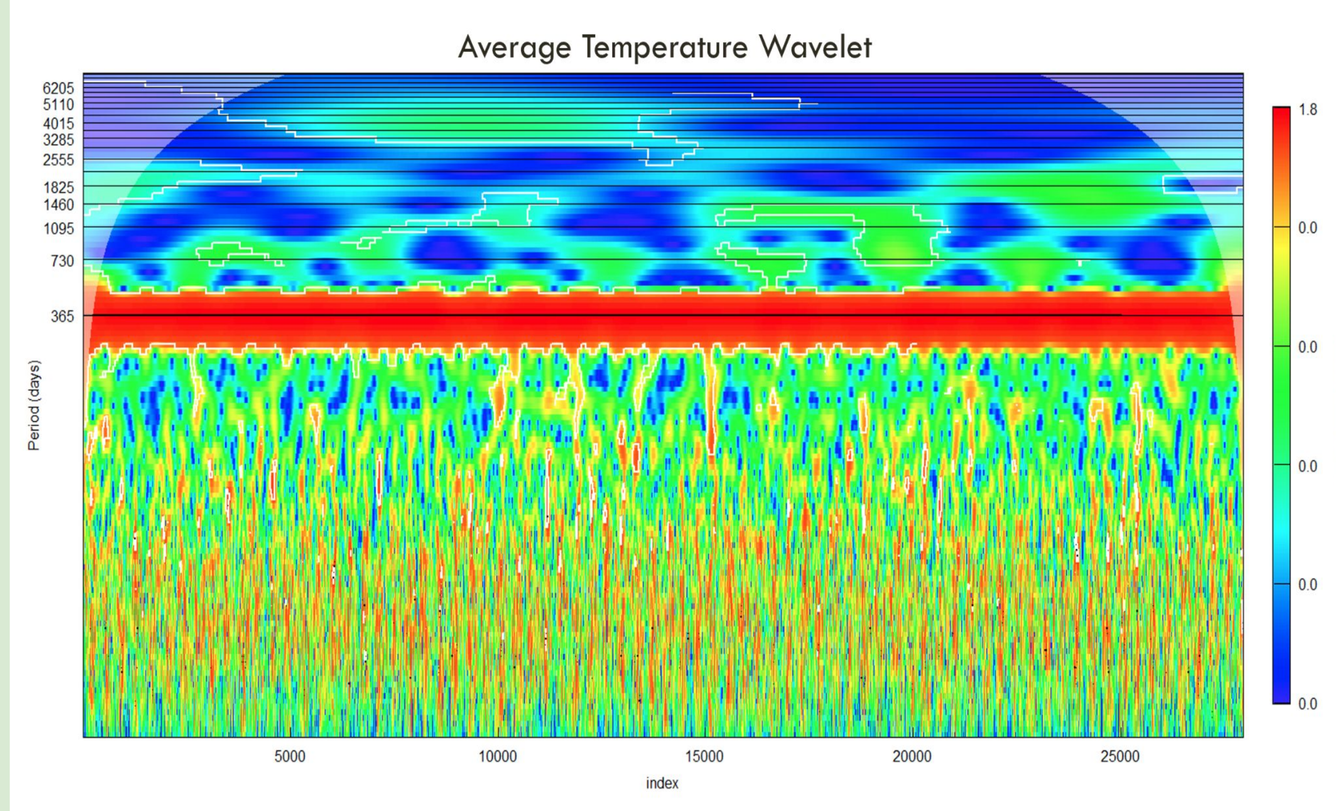
Sunspot Wavelet



Results From Sunspot Wavelet:

- The results reveal a clear **cyclical pattern with an approximate 11-year period**, consistent with the known solar sunspot cycle.
- **Peak intensity varies** between cycles, indicating that **not all solar maxima are equally strong**.
- The **11-year periodicity was confirmed** by the **wavelet power spectrum** and global wavelet analysis.
- Some cycles exhibit **higher amplitude peaks**, suggesting **greater sunspot activity** during those particular solar maxima.

Temperature Wavelet:



Results From Temperature Wavelet:

- **Temperature cycles show a consistent seasonal pattern**, dominated by a strong annual signal.
- The **wavelet power spectrum identifies a persistent 365-day period**, reflecting the expected yearly temperature cycle.
- This strong 365-day signal confirms the **regularity of seasonal temperature variations** over time.
- **Weaker signals around the 730-day range** may suggest potential multi-year modulations or variations in seasonal amplitude.
- **Further analysis with advanced noise-reduction techniques** could help detect subtle changes in yearly temperature peaks across decades.

Discussion: Sunspots' Effect on Average Temperature

- Wavelet analysis of sunspot data reveals a consistent 11-year solar cycle, characterized by periodic increases in solar activity. This finding was confirmed using the analyze.wavelet() function in the R WaveletComp package. The wavelet power spectrum clearly indicated dominant periodicities near 11 years, consistent with the well-documented Schwabe solar cycle.
- The corresponding temperature data for Islip, NY, when analyzed with the same wavelet technique, showed a strong and stable signal at the 365-day period, confirming the expected seasonal cycle. However, weaker signals above the 730-day (2-year) range were also present, suggesting possible multi-year modulations that may correlate with external forcings like sunspot activity.
- Time-series plots of normalized sunspot and temperature data (generated using scale() in R for standardization) visually suggest that years with high sunspot activity often correspond to slightly warmer average temperatures. This was particularly noticeable when lagging temperature behind sunspot peaks by several months, hinting at a delayed climatic response to solar forcing.
- Despite the presence of a general pattern, the relationship is not linear or deterministic. Certain high-sunspot years did not correspond with significant warming in Islip, and some warmer years occurred during solar minimums. This suggests that solar activity may be one of several contributing factors in regional temperature variability, with local atmospheric dynamics, oceanic cycles (e.g., ENSO), and anthropogenic influences also playing major roles.
- Extreme sunspot peaks (especially visible in the late 1950s, early 1980s, and around 2001) appear to correlate with slightly elevated normalized temperature anomalies, as visualized in the overlaid line plots created with ggplot2. However, these correlations weaken outside the main solar maxima, reinforcing the idea that the solar signal may be subtle compared to dominant seasonal and interannual temperature drivers.
- Lagged correlation analysis could enhance the interpretation of the sunspot-temperature link. This was hinted at by the phase offset between peaks in solar activity and shifts in temperature, but not fully explored in this preliminary analysis.

Conclusion:

- The Islip, NY temperature record (1948–present) reveals strong seasonality and potential weak multi-year cycles; the 11-year solar cycle is clearly evident in sunspot data but only marginally present in temperature records.
- There is a modest tendency for average temperatures to be slightly higher during or shortly after sunspot maxima, but this influence is not uniformly consistent and is often masked by dominant annual and short-term climate variability.
- Wavelet analysis proves useful in identifying both dominant and subtle periodic signals, allowing researchers to differentiate between consistent seasonal patterns and irregular long-term modulations like those potentially driven by solar activity.
- Further research will incorporate cross-wavelet coherence or lagged correlation techniques to better capture phase relationships between solar and temperature cycles. Additional noise-reduction techniques (e.g., smoothing or empirical mode decomposition) might enhance the detection of low-amplitude solar signals
- In summary, while solar activity, as measured by sunspots, appears to exert a secondary influence on Islip's average temperatures, the magnitude of this effect is small, and additional regional and global factors must be accounted for to accurately model temperature variability.

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Credit Authorship Contribution Statement:

Dominick Colucci: coding, poster, writing all sections.

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